OpenSSL Cheathseet

Basics

`openssl version -a`	Checking version
`openssl speed -multi 4 rsa`	Host fast it runs on the sysem four CPU cores and testing RSA algorithm
`openssl help`	Get basic help
`openssl rand -hex 20`	Generate 20 random bytes and show them on screen

Encoding/Decoding

`openssl base64 -in file.data`	Encoding a file using Base64
`echo -n "some text" | openssl base64`	Encoding some text using Base64
`openssl base64 -d -in encoded.data -out decoded.data`	Base64 decode a file with output to another file

Working with Hashes

`openssl list -digest-algorithms`	List digest algorithms available
`openssl dgst -sha256 file.data`	Hash a file using SHA256
`openssl dgst -binary -sha256 file.data`	Hash a file using SHA256 with its output in binary form (no output hex encoding). *No ASCII or encoded characters will be printed to the console, just pure bytes. You can append* `|xxd`
`echo -n "some text" | openssl dgst -sha3-512`	Hash text using SHA3-512
`openssl dgst -SHA384 -ma HMAX -macopt hexkey:369bd7d655`	Hash a file using SHA256 with its output in binary form (no output hex encoding). *No ASCII or encoded characters will be printed to the console, just pure bytes. You can append* `|xxd`

Reverting changes

`git rebase`	rebase the current branch on top of another specified branch
`git rebase -i [commit sha]`	start an interactive rebase
`git revert [commit sha]`	Create a new commit, reverting changes from the specified commit. It generates an **inversion** of changes.
`git checkout <path>`	discard changes in the **working tree**
`git restore [-W/--worktree] <path>`	discard changes in the **working tree**
`git restore -S/--staged <path>`	remove a file from a **staging area**
`git restore -SW <path>`	discard changes in the **working tree** and to the **staged** files
`git reset <path>`	remove a file from the **staging area**
`git reset [mode] HEAD^`	remove the latest **commit** from the current branch and: • `--soft` - keep file changes in the **working tree** and **stage** them; • `--mixed` - keep file changes; • `--keep` - reset only files which are different between current `HEAD` and the last commit • `--hard` - do **not** keep file changes

Tagging commits

`git tag`	list all tags
`git tag <name> [commit sha]`	create a tag reference named `name` for the current or specific commit
`git tag -a <name> -m <message>`	create an annotated tag with the given message

`git tag -d <name>`	delete the tag with the given name

Synchronizing repositories

`git fetch [remote]`	fetch changes from the **remote**, but not update tracking branches
`git fetch --prune [remote]`	delete remote refs that were removed from the **remote** repository
`git pull [remote]`	fetch changes from the **remote** and **merge** current branch with its upstream
`git pull -r/--rebase [remote]`	fetch changes from the **remote** and **rebase** current branch on top of the upstream
`git push -u [remote] [branch]`	push local branch to a **remote** repository and set its copy as an upstream